8 Programming

Before starting to program, you must choose your programming language (tick when complete). ☐

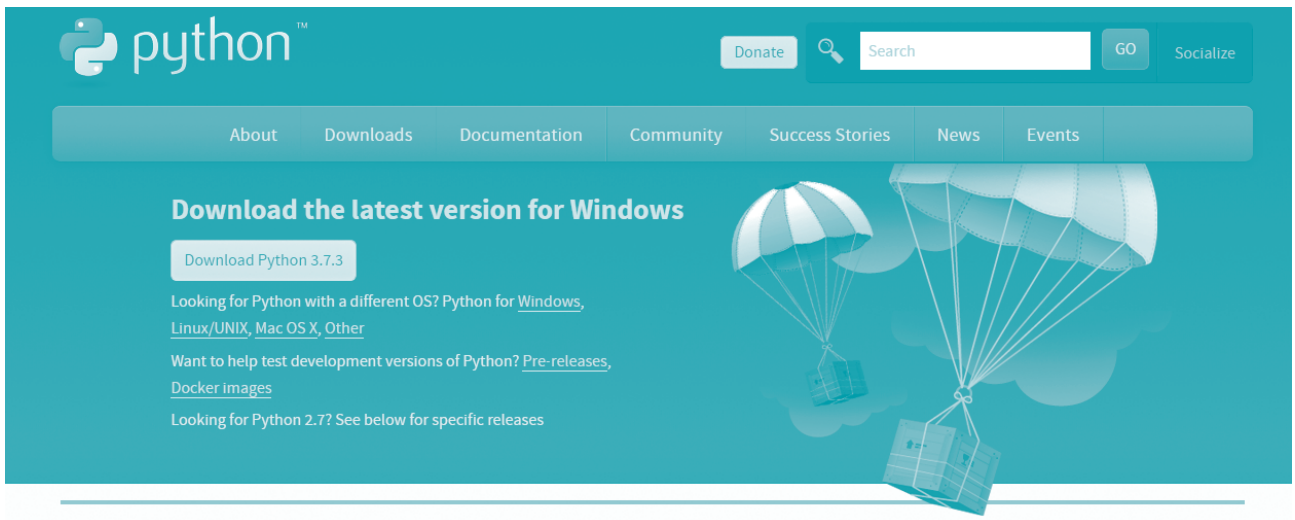
My choice of programming language is: .....

The integrated development environment (IDE)

If your school computers already have an integrated development environment (IDE) then you can use that to write your programs. Otherwise you will need to download and install a development environment to use. Here are some suggestions.

Python

https://www.python.org/downloads/



The download is free. Download and install it. The IDE, called IDLE, will provide an editing window:

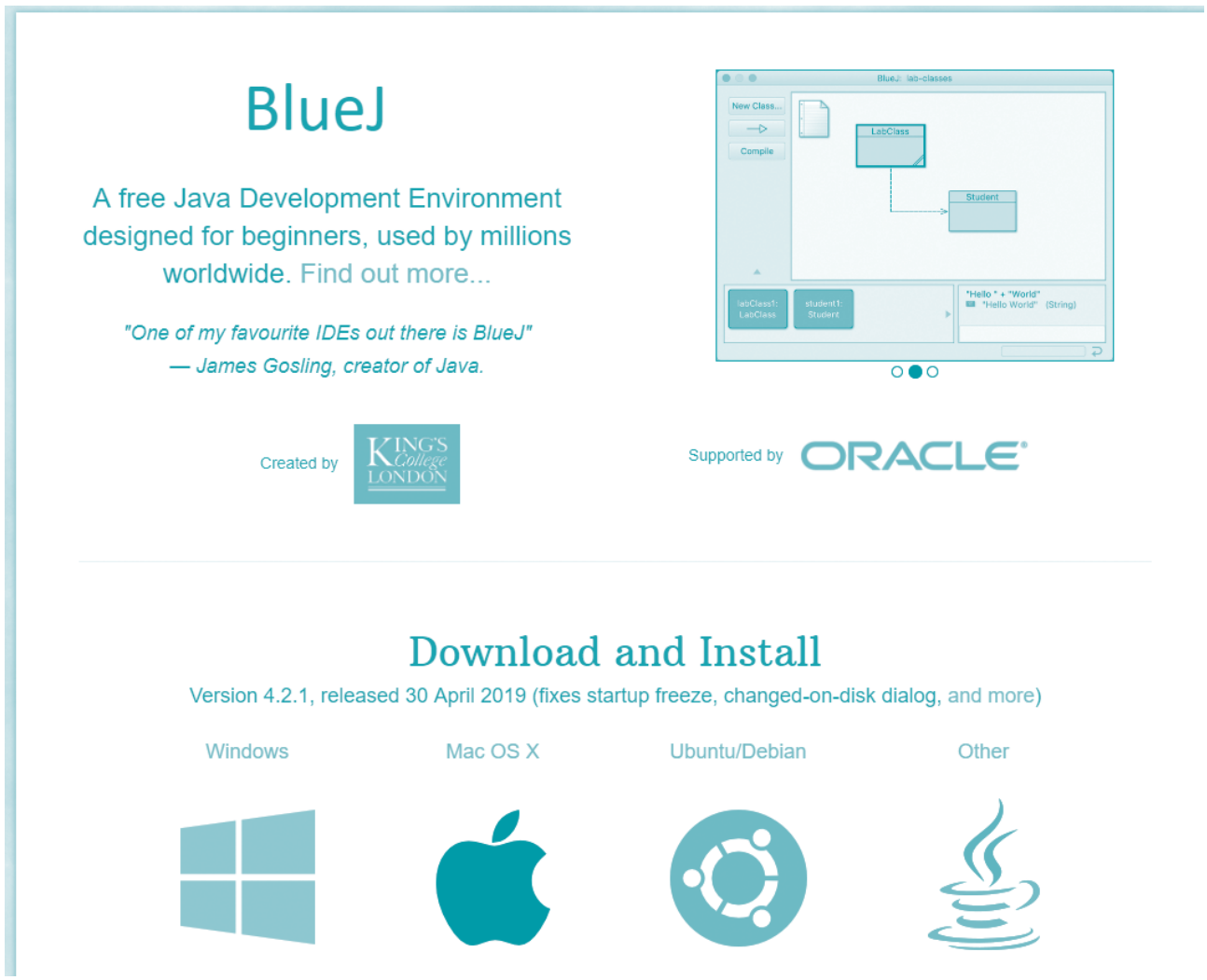


... and a runtime window:



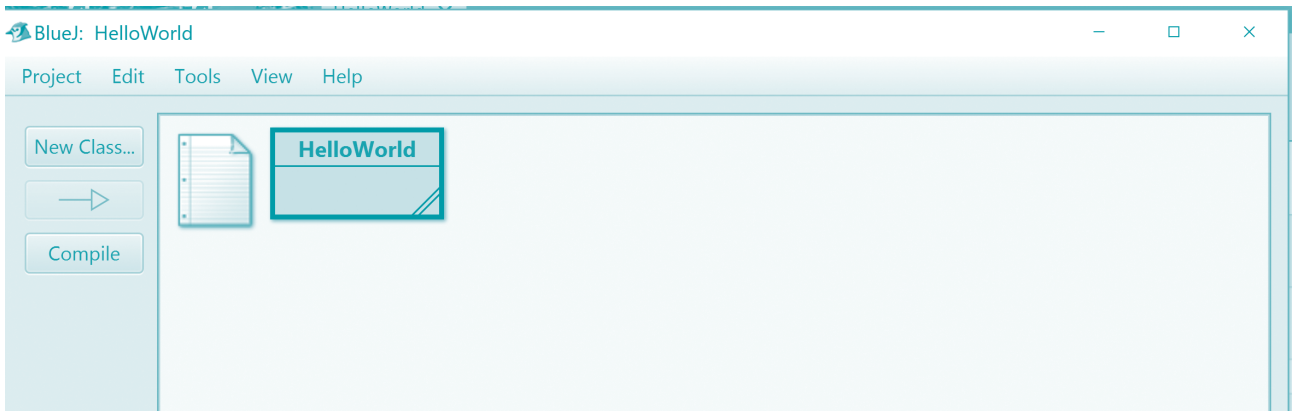
Java

https://www.bluej.org/



Photocopying prohibited

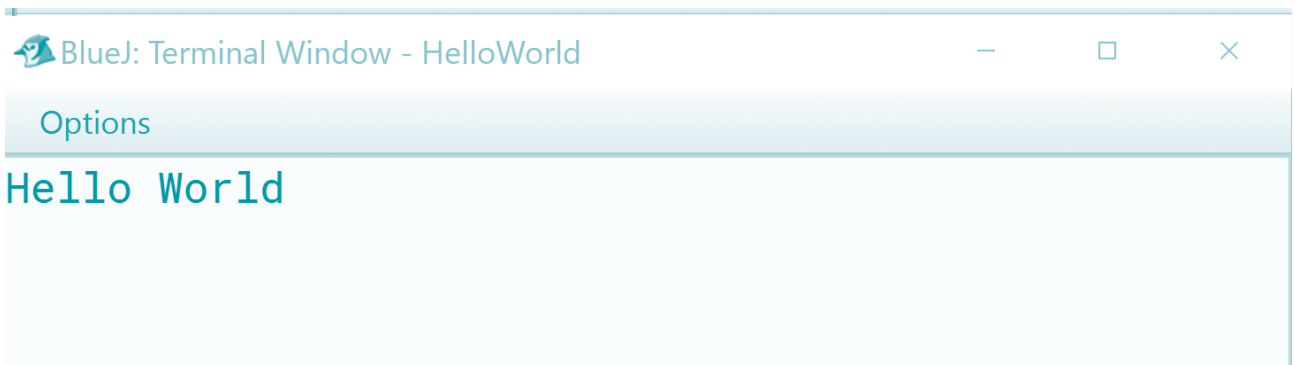
Download and install. It is free and will provide a project window:



... an editing window:



... and a terminal window to show the output from the program:

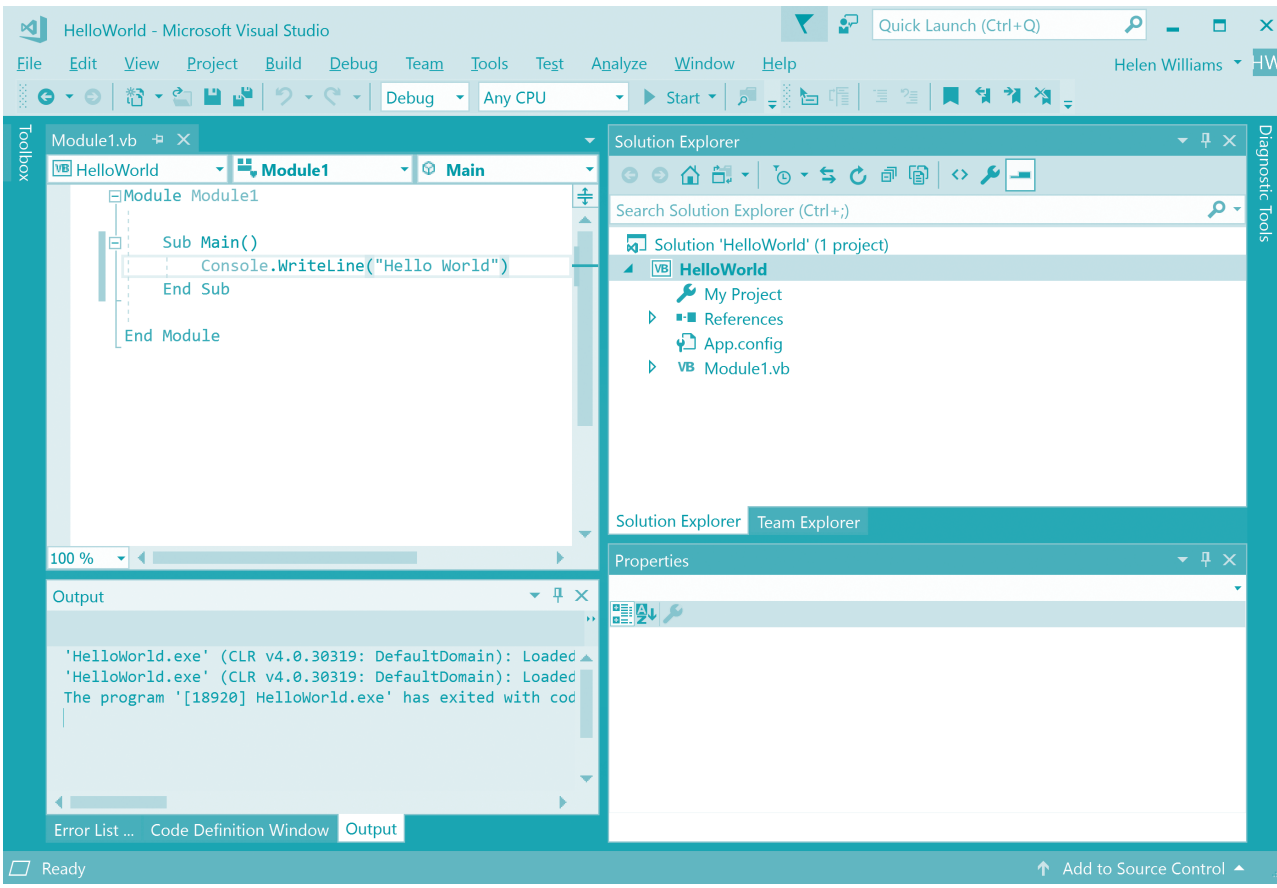


Visual Basic

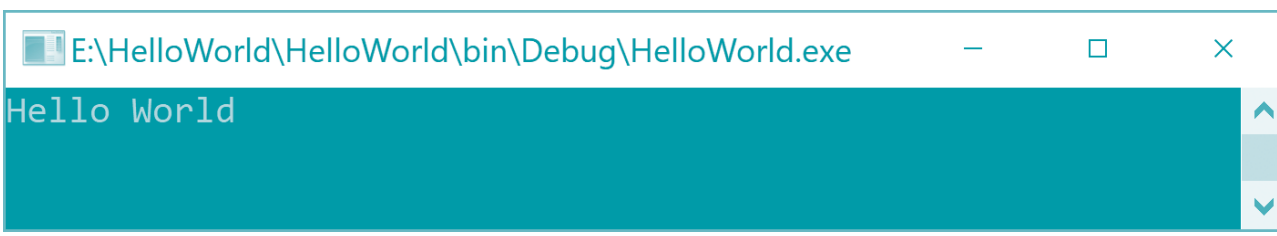
https://visualstudio.microsoft.com/vs/express/



Download and install the community version. It is free and will provide an editing window:



Programs written in console mode, once compiled, run at the command prompt:



Action: download and install the IDE for your chosen programming language (tick when complete). ☐

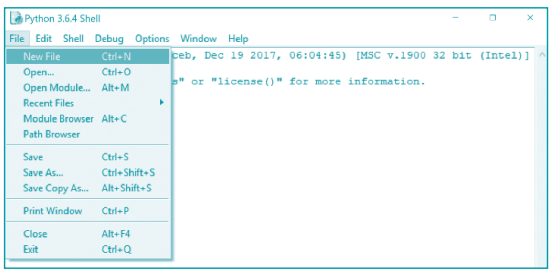
Now follow the instructions for your chosen programming language to write your first program.

Writing your first program in Python

Click on the IDE Icon to open Python.

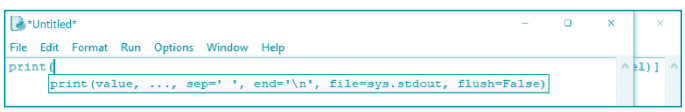


Choose **File>New File**.



The editing window will open.

Start typing your program. Help will appear automatically as you type.



To save your program choose **File>Save** or press **Ctrl S**.



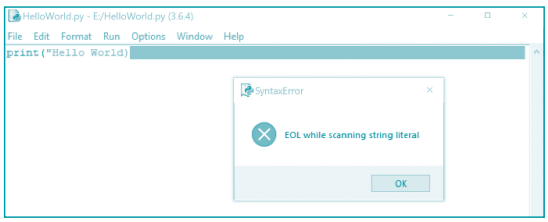
To run your program choose **Run>Run Module** or press **F5**.



If there are no syntax errors, then the program will run in the Python shell.



If there are syntax errors, then the IDE will show you where and provide an error message.



Action: write and run a Python program to output Hello World (tick when complete). ☐

Writing your first program in Java

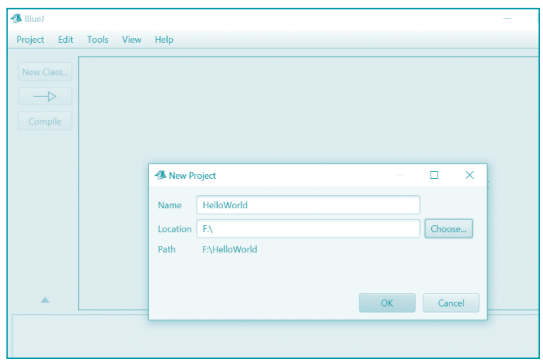
Click on the IDE Icon to open Java.



Choose **Project>New Project**.

Enter the file **Name** and **Location**.

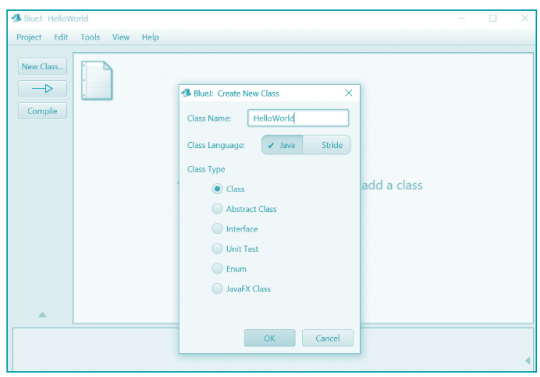
Then click on OK.



Choose **New Class**.

Give the project the **Class Name** HelloWorld. (THIS MUST BE THE SAME NAME AS THE BASE CLASS OF YOUR PROGRAM.)

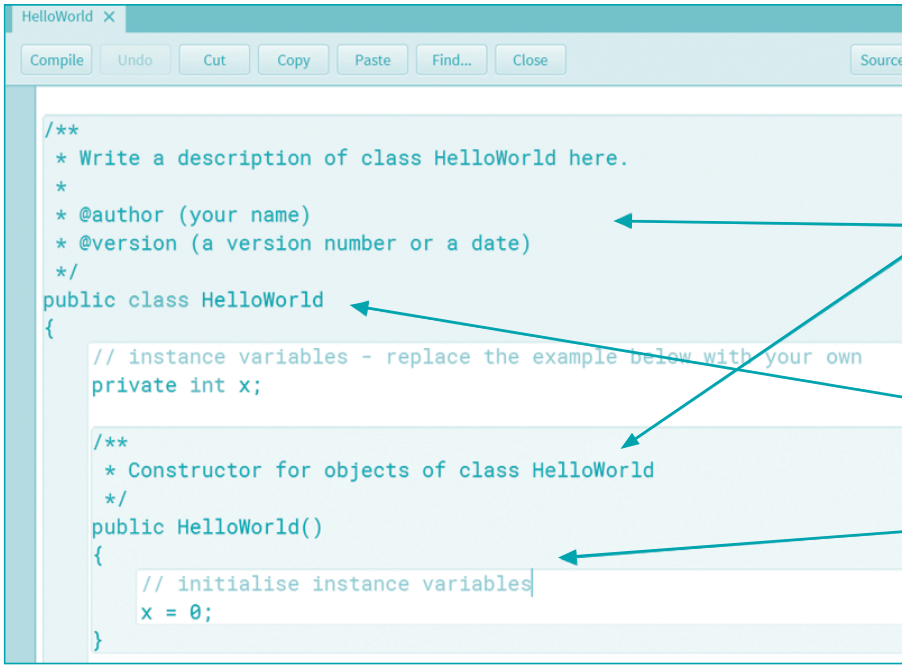
Choose OK and your new class is constructed.



The class icon appears on the screen.



Double click on the icon to edit your program.



The Edit window appears on the screen.

These are comments with hints for good programming.

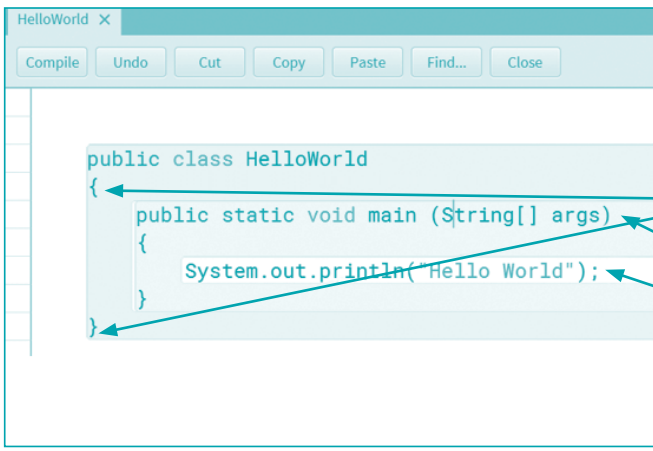
The HelloWorld class has been set up for you.

This is where you need to write your program.

Remove any lines not needed.

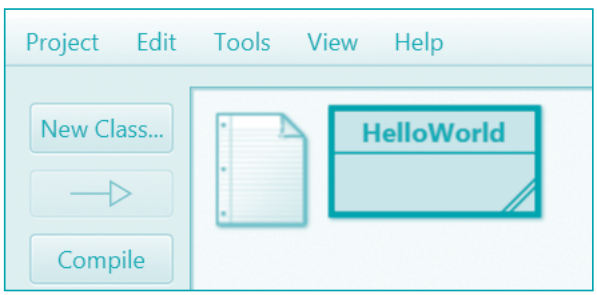
Type your program.

Once finished, press the **Compile** button.

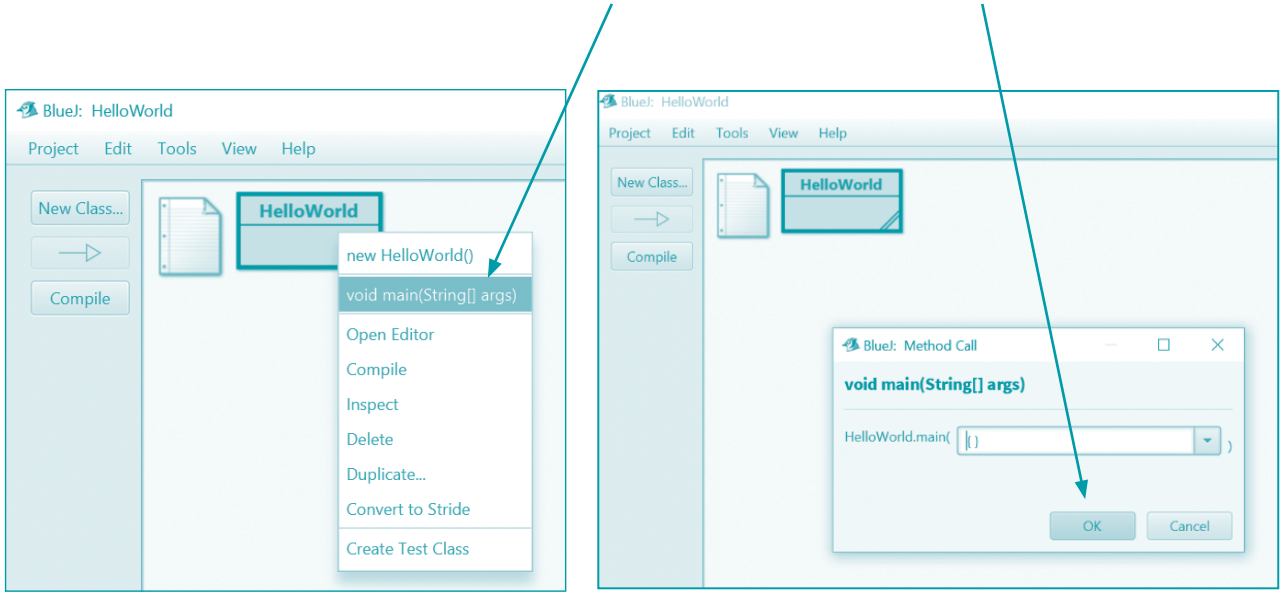


REMEMBER:
There must be matching braces at the start and end of every class.
There must be a main program.
There must be a semi-colon at the end of every programming statement.

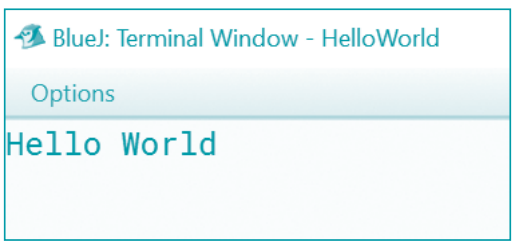
If the program has no syntax errors the class icon appears on the screen without any hatching.



Right click on the icon then choose **void main(String[] args)**. Then click on OK.



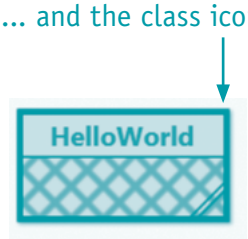
The program will run in a terminal window.



If there are syntax errors, then the IDE will show you where and provide an error message.



... and the class icon appears on the screen with hatching.



Action: write and run a Java program to output Hello World (tick when complete). ☐

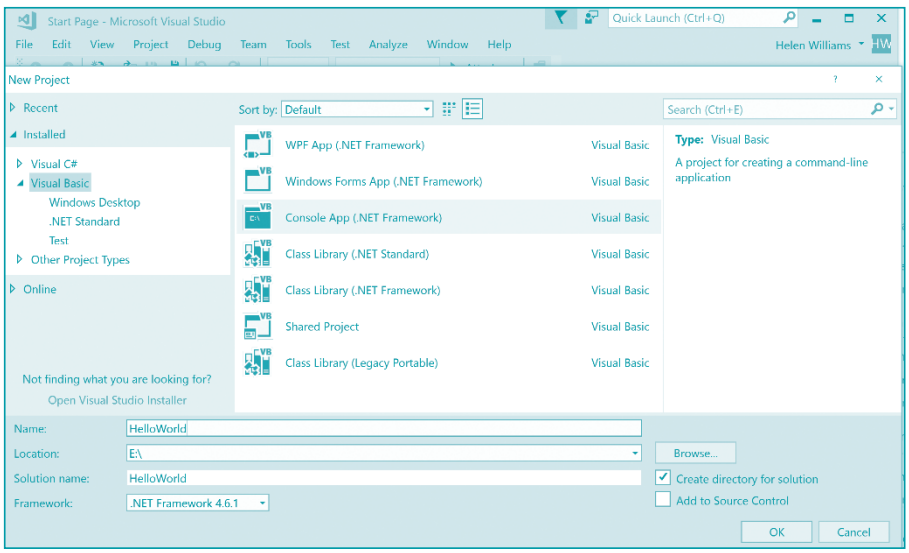
Writing your first program in Visual Basic

Click on the IDE Icon to open Visual Basic.

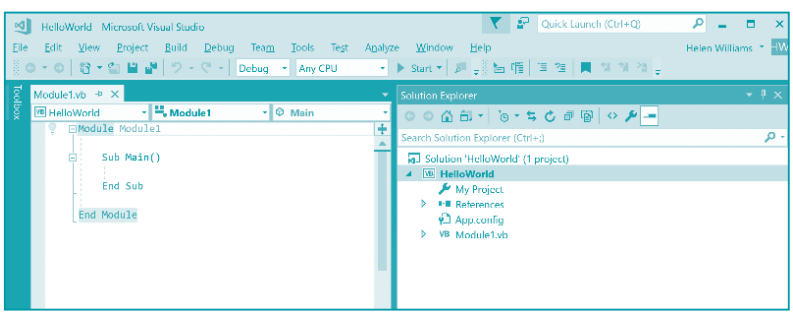
Choose **File>New>Project>Console App**.

Make sure the language selected is Visual Basic.

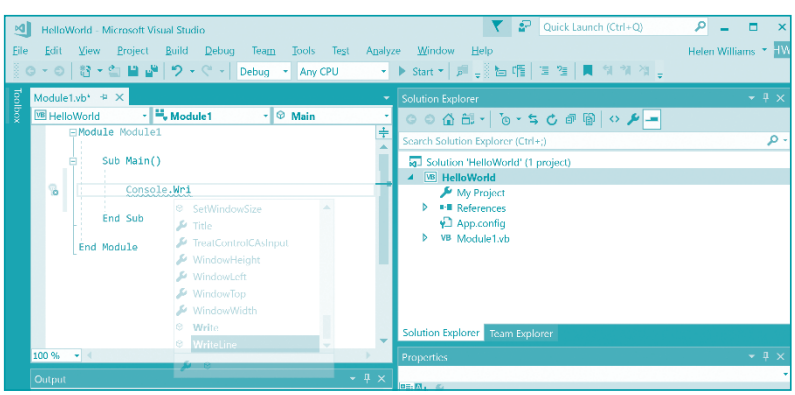
Then click on OK.



The editing window will appear with a skeleton program ready for you to add your code.

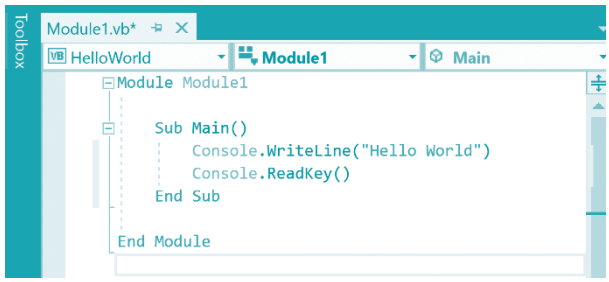


Suggestions will appear to help you write your code. Press **Enter** to accept a highlighted suggestion.

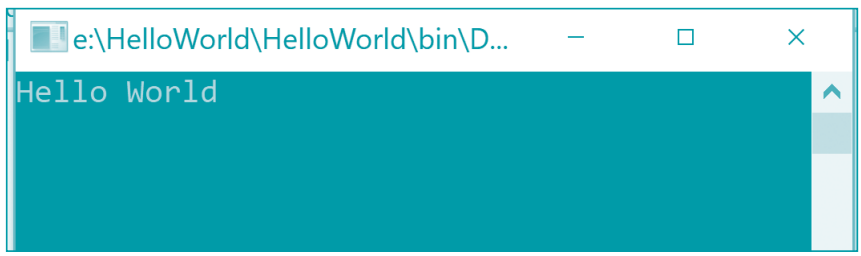


Your program will run in a command line window. The **Console.ReadKey()** will ensure that the output remains on screen.

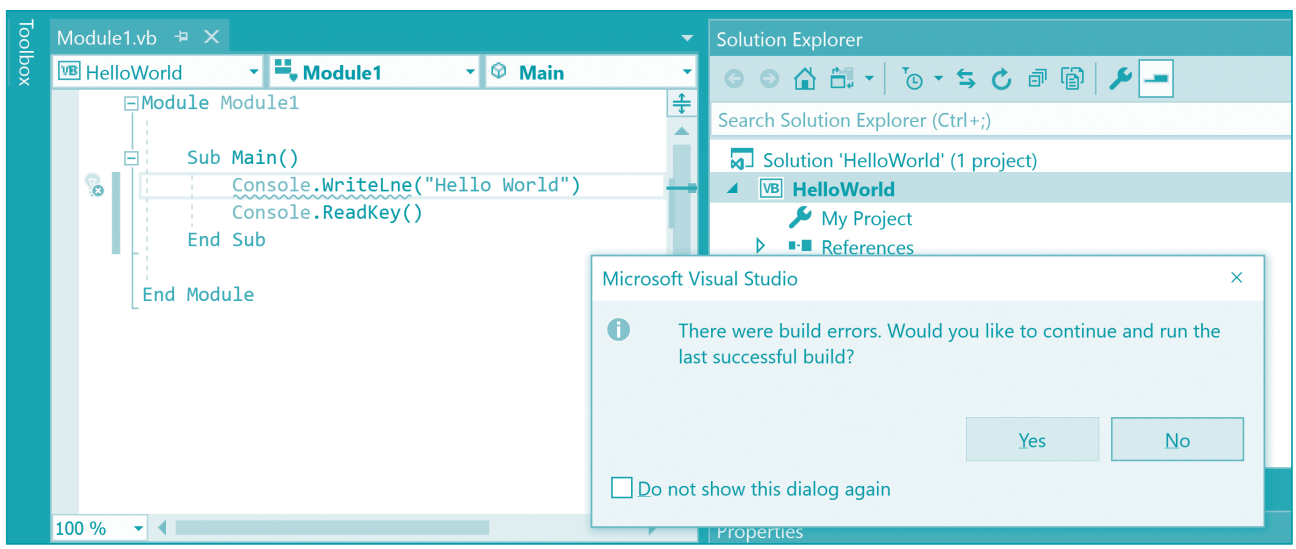
To run your program press **F5**.



This window will remain on screen until you press a key:



If there are syntax errors, then the IDE will show you where and provide an error message.



Action: write and run a Visual Basic program to output Hello World (tick when complete). ☐

Getting started

There is plenty of information about getting started using an IDE. Remember you will be writing programs in console mode. Here are some examples.

Python using IDLE

http://web.mit.edu/6.s189/www/handouts/GettingStarted.html
https://sites.physics.utoronto.ca/comp-physics/manual/tutorial-part-1-first-steps-with-idle-and-python

Java using BlueJ

https://www.cs.utexas.edu/~scottm/cs307/handouts/BlueJProjectInstructions.html
http://cs.carleton.edu/faculty/jondich/courses/cs117_w02/labs/lab1.html

Visual Basic using Visual Studio

https://docs.microsoft.com/en-us/visualstudio/ide/quickstart-visual-basic-console?view=vs-2019
https://www.ict.social/vbnet/basics/visual-studio-and-your-first-vbnet-console-application

Action: check out one of the links for your chosen programming language (tick when complete). ☐

Action: find and watch a YouTube video for using your chosen programming language in console mode (tick when complete). ☐

1 Identify and describe **three** basic data types.

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2 .....

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3 .....

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2 a In your chosen programming language write a prompt and input statement(s) to enter **three** whole numbers.

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b In your chosen programming language write statement(s) to assign the values, 10, 20 and 30 to **three** variables.

c In your chosen programming language write a message and an output statement to display **three** whole numbers that have been stored.

d In your chosen programming language, using your answers to **a**, **b** and **c** above, write and test a program that assigns **three** user-input numbers to **three** variables and displays the values; then assigns numbers to the same three variables and displays the values. You may need to declare the variables used to store your numbers. Write or paste a copy of your program here.

3 a In your chosen programming language write a statement to only accept a parcel between 0.5 and 5.0 kilograms.

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b In your chosen programming language write a statement to only accept a menu choice that is 1, 2, 3, 4 or 5.

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c In your chosen programming language write and test a program that:

i accepts parcels between 0.5 and 5.0 kilograms in weight

ii displays a menu of options for the cost of sending the parcel

iii calculates and outputs the cost
- option 1 weight times $10 plus $1
- option 2 weight times $10
- option 3 $5
- option 4 $4
- option 5 $3

Options	
1	Guaranteed next day delivery before noon
2	Guaranteed next day delivery
3	24-hour delivery
4	48-hour delivery
5	3–5 days delivery

Write or paste a copy of your program here.

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d Test your program with the following test data and complete this table.

Test data	Expected output	Actual output
0.4		
5.1		
–1		
0.5 Option 1		
0.5 Option 2		
0.5 Option 3		
5 Option 1		
5 Option 2		
5 Option 4		
1 Option 5		

4 Describe each program concept and give an example in a programming language.

a Counting:

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b Iteration:

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c Selection:

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d Sequence:

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e String handling:

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5 Put the following pseudocode statements in the correct order for this algorithm. Make sure that you indent each statement correctly.

```
// Algorithm to input ten positive numbers and total them, output the total and then
// average
A ← 0
A ← A + C
B ← 0
B ← B + 1
D = A / B
DECLARE A, B, C, D : INTEGER
DECLARE D : REAL
INPUT C
OUTPUT A, D
OUTPUT "Please enter a positive number "
REPEAT
REPEAT
UNTIL B = 10
UNTIL C > 0
```

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6 a In your chosen programming language write a program for an improved algorithm for Question **5**. Your program must use:

- meaningful variable names
- appropriate prompts and messages
- relevant comments.

Write or paste a copy of your program here.

b Test your program with this test data:

4, 3, 7, 6 ,0, –3, 9, 1, 8, 5, 2, 5

Show your results in this trace table.

7 **a** Choose two different types of loop structure.

For each one, write and test a program to output the names of eight students that have been stored in the array `Student[0:7]`. Use the variable `Counter` for the iteration.

b Extend one of your programs to work for any number of students, outputting the list of students and number of students in the array.

Write or paste a copy of your program here.

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8 a Write pseudocode statements to perform the following string handling operations on the variable `MyString`.

Find the length .....

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Convert to upper case .....

.....

Convert to lower case .....

.....

Find the first character .....

.....

b In your chosen programming language write a program to perform the four string operations from part **a**. Test your program with the string `"Test String"` stored in the variable `MyString`. Write or paste a copy of your program here.

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9 a This pseudocode statement performs a calculation:

```
A ← ((B + C – D * E) ^ F) / G
```

Write this statement in your chosen programming language.

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b This pseudocode statement uses a nested `IF` and performs a logical operation:

```
IF (A = B) OR (C <> D)
 THEN
   A ← 0
 ELSE
   IF (A > B) AND (C = D)
     THEN
       B ← 0
   ENDIF
ENDIF
```

Write this statement in your chosen programming language.

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c Write pseudocode statements to use a nested `FOR .. NEXT` loop to output the numbers 1 to 10 twenty times.

d Write and test a program to use a nested `FOR .. NEXT` loop to output the numbers 1 to 10 twenty times. Write or paste a copy of your program here.

10 Use the following list of words and phrases to complete the paragraph below.

Each word can be used once, more than once or not at all.

- called
- code
- constant
- defined
- functions
- global
- local
- parameters
- procedures
- program
- return
- value

Tasks that are repeated many times in an algorithm make use of ..... and ..... Use of these can reduce the size of a ..... .

Procedures and functions are ..... once and ..... many times. They can be written with and without ..... . They are the variables that store the values passed to a procedure or function.

Functions always ..... a ..... and the value can be used on the right-hand side of an assignment statement.

A variable that can be used in any part of a program is called a ..... variable.

A variable that is declared within a procedure or function is called a ..... variable.

11 a Describe the purpose of a library routine.

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b Identify **four** library routines that you need to be able to use for IGCSE.

1 .....

2 .....

3 .....

4 .....

c Write and test a program that uses library routines to create two random integers with values between 10 and 20, finds the quotient and remainder of the first integer divided by the second, and outputs all four values with suitable messages. Write or paste a copy of your program here.

12 a Explain the meaning of the following terminology.

Array: .....

Array dimension: .....

Array index: .....

b Write a procedure in pseudocode to display the contents of a three by three array that stores the current state of a noughts and crosses game. The display should look just like a noughts and crosses game, using 'X's and 'O's on three separate lines.

c Write a program that uses a procedure to display the contents of a three by three array that stores the current state of a noughts and crosses game. Test the program with this data. Each row must start on a new line. Write or paste a copy of your program here.

Here is an example of what might be displayed on screen:

```
O
  X
  X  O
```

13 a Explain the purpose of using files to store data.

b i Write pseudocode statements to store the word 'Test' in the file 'MyFile.txt'.

ii Write pseudocode statements to read and display the word stored in the file 'MyFile.txt'.

c Write a program to store the word 'Test' in the file 'MyFile.txt', then read and display the word stored in the file 'MyFile.txt'. Write or paste a copy of your program here.

14 a Write a program to ask a user if they want to:

1 Enter a new password
2 Check their password
3 Change their password
4 Quit

The program should call an appropriately named procedure or function for each option. It should also include a loop that only stops if option 4 is selected.

Write or paste a copy of your program here.

b Write a procedure or function for option 1 from the program in part **a** to input a word to use for a password. It should check the following about the word:

- It is between 10 and 20 characters in length.
- It does not contain any spaces.
- It contains at least one uppercase letter.
- It contains at least one digit between 0 and 9.

The procedure or function should output suitable error messages if the password does not meet each of these conditions. Write or paste a copy of your procedure/function here.

Test your program with the following test data and complete this table.

Test data	Expected output	Actual output
Password		
Password99		
password99		
Password99!		
ExtraLargePassword9999999		
2 1		

c Extend your code from part **b** to store a valid password in the file 'MyPassword.txt'.

Write or paste a copy of the extension to your procedure/function here.

d Write another procedure or function for option 2 from the program in part **a** to input a password and check that it matches the password stored in 'MyPassword.txt'. Write or paste a copy of your procedure/function here.

e Write another procedure or function for option 3 from the program in part **a**. It should ask the user to enter the current password and check that this is correct (i.e. matches the one stored in 'MyPassword.txt') before allowing them to change their password. The new password should match all of the requirements of a password listed in part **b**. (HINT: It is possible to call one procedure or function from within another procedure or function.) Write or paste a copy of your procedure/function here.

15 Write and test a program that uses a two-dimensional array, `Game[]` to store the moves in a noughts and crosses game. The program should meet the following requirements:

- Start each game with an empty array.
- Allow two players to input their moves in turn; the contents of the array are displayed before and after every move.
- One player can input O and the other input X; no other moves are allowed, and no move can use a space in the array already occupied.
- After every move the program should check for a completed line of three Os or three Xs, and if found output the winner of the game.
- Use procedures and parameters.
- Include comments to explain how your code works.